ANSWER KEY · 7TH GRADE LIFE SCIENCE

Final Exam

Complete answer key with worked solutions for all 80 questions. Multiple choice answers (Q1-65) are highlighted in green and tagged "✓ ANSWER." Fill-in answers (Q66-75) and free-response model answers (Q76-80) are shown in full beneath each prompt.

MS. MADEWELL · ROOM 501 · ALA VISTANCIA

65 MC Answers 10 Fill-in Answers 5 FRQ Model Solutions

QUICK REFERENCE · MULTIPLE CHOICE ANSWERS (1-65)

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9
C	A	C	C	C	C	A	B	B
Q10	Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18
C	C	B	C	C	D	B	D	C
Q19	Q20	Q21	Q22	Q23	Q24	Q25	Q26	Q27
A	B	A	C	B	C	B	A	C
Q28	Q29	Q30	Q31	Q32	Q33	Q34	Q35	Q36
B	B	B	A	D	B	C	B	B
Q37	Q38	Q39	Q40	Q41	Q42	Q43	Q44	Q45
C	A	C	C	B	C	B	D	B
Q46	Q47	Q48	Q49	Q50	Q51	Q52	Q53	Q54
B	C	C	B	B	C	C	B	B

Q55	Q56	Q57	Q58	Q59	Q60	Q61	Q62	Q63
C	B	C	B	B	B	A	C	B

Q64	Q65
B	C

SECTION A · MULTIPLE CHOICE (1-65)

Question 1

NON-CONTACT FORCES

Which of the following is a **non-contact** force?

A Friction

B Tension in a rope

C Gravity ✓ ANSWER

D A push from your hand

SOLUTION

A non-contact force can act on an object without touching it. Friction, tension, and a push all require physical contact. Gravity pulls between any two objects with mass even across empty space — the only non-contact force in the list.

Question 2

NON-CONTACT FORCES

Two magnets are placed near each other and they push apart. This means:

A Like poles are facing each other (N-N or S-S) ✓ ANSWER

B Opposite poles are facing each other

C One of the magnets is broken

D The magnets have no charge

SOLUTION

Magnetic rule: like poles repel, opposite poles attract. If the magnets push apart, that's repulsion — so like poles are facing each other.

Question 3

NON-CONTACT FORCES

As the distance between two objects with mass **increases**, the gravitational force between them:

A Stays exactly the same

B Increases

C Decreases ✓ ANSWER

D Becomes negative

SOLUTION

Gravity gets weaker as objects get farther apart. It never reaches zero (it just keeps shrinking) and gravity is never negative — it only pulls. So gravity decreases.

Question 4

NON-CONTACT FORCES

A balloon is rubbed on someone's hair, then sticks to a wall. The non-contact force responsible is:

A Magnetism

B Gravity

C Static electricity ✓ ANSWER

D Friction

SOLUTION

Rubbing the balloon transfers electrons onto it, giving it a negative charge. The charged balloon induces an opposite charge in the wall, and opposite charges attract. This is static electricity.

Question 5

NON-CONTACT FORCES

When an object becomes **negatively** charged, what particles have moved onto it?

A Protons

B Neutrons

C Electrons ✓ ANSWER

D Whole atoms

SOLUTION

Electrons are the particles that move during charging — protons stay locked in the nucleus, and neutrons have no charge. Gaining electrons → object becomes negative.

Question 6

NON-CONTACT FORCES

Gravity on the Moon is about 1/6 of Earth's. An astronaut's **mass** on the Moon, compared to Earth, is:

A 1/6 as much

B 6 times greater

C Exactly the same ✓ ANSWER

D Zero

SOLUTION

Mass is the amount of matter in an object — it doesn't change with location. Weight (the pull of gravity) IS smaller on the Moon, but mass stays exactly the same.

Question 7

NON-CONTACT FORCES

The region of space around a magnet where it can exert a force on another magnet is called the:

- A Magnetic field ✓ ANSWER
- B Charge cloud
- C Gravity well
- D Static zone

SOLUTION

Every non-contact force has a *field* — the area in which the force acts. For magnets, that's the magnetic field. (Gravity has a gravitational field; charged objects have an electric field.)

Question 8

NON-CONTACT FORCES

Which best describes **weight**?

- A The amount of matter in an object
- B The pull of gravity on an object's mass ✓ ANSWER
- C The volume of an object
- D The speed at which an object falls

SOLUTION

Weight = mass × gravity. It's a *force*, measured in newtons. "Amount of matter" is mass; "space taken up" is volume. Weight is the gravitational

pull on an object's mass.

Question 9

NON-CONTACT FORCES

Object A has a charge of **+2**. Object B has a charge of **−3**. The electric force between them will be:

A Repulsive (push apart)

B Attractive (pull together) ✓ ANSWER

C Zero

D Magnetic only

SOLUTION

Charge rule: like charges repel, opposite charges attract. One is positive (+2), one is negative (−3) — opposite signs — so the force is attractive.

Question 10

NON-CONTACT FORCES

If you snap a bar magnet exactly in half, what happens?

A You get one piece with N pole and one piece with S pole, separated

B Both pieces lose their magnetism

C You get two new magnets, each with its own N pole and S pole ✓ ANSWER

D One piece keeps both poles, the other has none

SOLUTION

A magnet's poles come from the alignment of tiny magnetic regions inside it. No matter how many times you cut it, each piece becomes a new magnet with its own N and S pole. Magnets cannot have just one pole.

Question 11

CELLS

The basic unit of all living things is the:

A Atom

B Tissue

C Cell ✓ ANSWER

D Organ

SOLUTION

From cell theory: the cell is the smallest unit that is alive. Atoms are non-living building blocks; tissues and organs are made up of many cells.

Question 12

CELLS

Which organelle is often called the **"powerhouse of the cell"** because it produces most of the cell's energy?

A Nucleus

B Mitochondria ✓ ANSWER

C Ribosome

D Vacuole

SOLUTION

Mitochondria perform *cellular respiration*, breaking down glucose to release ATP (chemical energy). That's why they're nicknamed the powerhouse.

Question 13

CELLS

Which structure is found in **plant** cells but NOT in animal cells?

A Nucleus

B Mitochondria

C Cell wall ✓ ANSWER

D Ribosome

SOLUTION

All four are in plant cells, but only the cell wall (and chloroplasts and the large central vacuole) are missing in animal cells. The cell wall gives plants rigid structure.

Question 14

CELLS

Photosynthesis takes place primarily in which organelle?

A Mitochondria

B Nucleus

C Chloroplast ✓ ANSWER

D Lysosome

SOLUTION

Chloroplasts contain chlorophyll, the green pigment that captures sunlight. They turn CO_2 + water + light → glucose + O_2. Only plants and some other organisms have them.

Question 15

CELLS

Which statement is **NOT** part of the cell theory?

A All living things are made of one or more cells

B The cell is the basic unit of life

C All cells come from other cells

D Every cell must contain a nucleus ✓ ANSWER

SOLUTION

A, B, and C are the three parts of cell theory. D is FALSE — prokaryotic cells (like bacteria) have DNA but NO membrane-bound nucleus, and they're still cells.

Question 16

CELLS

The **"control center"** of a eukaryotic cell, where DNA is stored, is the:

A Mitochondria

B Nucleus ✓ ANSWER

C Vacuole

D Cell membrane

SOLUTION

The nucleus contains the cell's DNA — the instructions that direct everything the cell does. It controls cell growth, division, and protein production.

Question 17

CELLS

Prokaryotic cells (such as bacteria) do **NOT** have which of the following?

A DNA

B A cell membrane

C Cytoplasm

D A membrane-bound nucleus ANSWER

SOLUTION

Prokaryotes DO have DNA, a cell membrane, and cytoplasm — but their DNA floats freely in the cytoplasm. Only eukaryotes (plants, animals, fungi, protists) wrap their DNA in a membrane-bound nucleus.

Question 18

CELLS

The thin barrier that controls what enters and leaves a cell is the:

A Cell wall

B Nucleus

C Cell membrane ✓ ANSWER

D Vacuole

SOLUTION

The cell membrane is "selectively permeable" — it lets some things in (food, oxygen) and keeps other things out, while sending wastes out. Plants have a cell membrane *and* a cell wall outside it.

Question 19

CELLS

Which list shows the levels of organization from **smallest to largest**?

- **A** Cell → tissue → organ → organ system ✓ ANSWER
- B Tissue → cell → organ → organ system
- C Organ → tissue → cell → organ system
- D Organism → organ → tissue → cell

SOLUTION

Smallest to largest: cell → tissue → organ → organ system → organism. Many cells of the same kind form a tissue; tissues form organs; organs form systems.

Question 20

CELLS

Which equation best summarizes **photosynthesis**?

- A Glucose + O_2 → CO_2 + H_2O + energy
- **B** CO_2 + H_2O + light → glucose + O_2 ✓ ANSWER
- C Water → hydrogen + oxygen
- D ATP → ADP + heat

SOLUTION

Plants take in CO_2 from the air and water from the soil, use sunlight as energy, and produce glucose (food) + oxygen. Choice A is the reverse —

that's *cellular respiration*.

Question 21

CELLS

Cellular respiration releases usable energy by breaking down which molecule?

- A Glucose ✓ ANSWER
- B DNA
- C Water
- D Cell walls

SOLUTION

In the mitochondria: glucose + oxygen → CO_2 + H_2O + ATP. The chemical bonds in glucose store energy, and respiration unlocks that energy as ATP for the cell to use.

Question 22

HUMAN BODY SYSTEMS

Which body system pumps blood through the body?

- A Respiratory
- B Nervous
- C Circulatory ✓ ANSWER
- D Digestive

SOLUTION

The circulatory system = heart + blood + blood vessels (arteries, veins, capillaries). Its job is to pump and distribute blood to every cell in the body.

Question 23

HUMAN BODY SYSTEMS

The **diaphragm** is a major muscle for which body system?

A Skeletal

B Respiratory ✓ ANSWER

C Digestive

D Excretory

SOLUTION

The diaphragm is a dome-shaped muscle below the lungs. When it contracts (flattens), the chest expands and air rushes IN; when it relaxes, air pushes OUT. It powers breathing for the respiratory system.

Question 24

HUMAN BODY SYSTEMS

Where does most **chemical digestion** and absorption of nutrients occur?

A Stomach

B Esophagus

C Small intestine ✓ ANSWER

D Large intestine

SOLUTION

The stomach mainly mashes food and starts breaking down proteins. Most chemical digestion (with help from enzymes from the pancreas and liver) AND nutrient absorption happens in the small intestine, which is lined with tiny villi to maximize surface area.

Question 25

HUMAN BODY SYSTEMS

Which system is made up of the brain, spinal cord, and nerves and sends signals throughout the body?

A Endocrine

B Nervous ✓ ANSWER

C Skeletal

D Muscular

SOLUTION

The nervous system uses fast electrical signals through neurons. The endocrine system also sends messages, but using slower chemical hormones, not nerves.

Question 26

HUMAN BODY SYSTEMS

The main job of **red blood cells** is to carry:

A Oxygen ✓ ANSWER

B Hormones only

C Solid wastes

D Glucose only

SOLUTION

Red blood cells contain hemoglobin, an iron-rich protein that grabs oxygen at the lungs and releases it at the body's tissues. (They also help carry some CO_2 back to the lungs.)

Question 27

HUMAN BODY SYSTEMS

Bones meet other bones at structures called:

A Tissues

B Organs

C Joints ✓ ANSWER

D Tendons

SOLUTION

Joints are where bones meet (knees, elbows, knuckles). Tendons connect *muscle to bone*; ligaments connect *bone to bone*.

Question 28

HUMAN BODY SYSTEMS

The kidneys, which filter wastes from the blood, are part of which system?

A Digestive

B Excretory ✓ ANSWER

C Respiratory

D Endocrine

SOLUTION

The excretory system (also called urinary system) includes kidneys, ureters, bladder, and urethra. Kidneys filter wastes — especially urea — out of the blood and form urine.

Question 29

HUMAN BODY SYSTEMS

Hormones are chemical messengers released by which system?

A Nervous

B Endocrine ✓ ANSWER

C Skeletal

D Integumentary

SOLUTION

The endocrine system includes glands like the pituitary, thyroid, and adrenal glands. They release hormones into the blood to slowly regulate growth, mood, energy, and more.

Question 30

HUMAN BODY SYSTEMS

The respiratory and circulatory systems work **together** to:

A Digest food into nutrients

B Exchange oxygen and carbon dioxide between air and blood ✓ ANSWER

C Filter and remove urine

D Send electrical signals to the brain

SOLUTION

In the lungs' *alveoli*, oxygen passes into the blood while CO_2 leaves it. The respiratory system handles air-to-blood exchange; the circulatory system carries those gases between lungs and cells. Together: gas exchange.

Question 31

HUMAN BODY SYSTEMS

Skin, hair, and nails make up the:

A Integumentary system ✓ ANSWER

B Endocrine system

C Skeletal system

D Lymphatic system

SOLUTION

The integumentary system is the body's outer covering. It protects against pathogens, regulates temperature (sweat), and senses touch.

Question 32

HUMAN BODY SYSTEMS

Which is NOT a function of the skeletal system?

A Supports the body

B Protects organs like the brain and lungs

C Produces blood cells in bone marrow

D Pumps blood ✓ ANSWER

SOLUTION

Bones DO support, protect, and produce blood cells (in red bone marrow). Pumping blood is the heart's job — that's the *circulatory* system, not skeletal.

Question 33

WEATHER & BIOMES

What is the difference between **weather** and **climate**?

A Climate is short-term, weather is long-term

B Weather is day-to-day; climate is the long-term average over many years ✓ ANSWER

C They mean exactly the same thing

D Climate only describes deserts

SOLUTION

Weather = what's happening RIGHT NOW or today (rain, heat, snow). Climate = the long-term pattern over decades. "It's snowing today" is weather.

"Phoenix is hot and dry" is climate.

Question 34

WEATHER & BIOMES

A rapidly rotating column of air that touches the ground is a:

A Hurricane

B Blizzard

C Tornado ✓ ANSWER

D Drought

SOLUTION

Tornadoes form during thunderstorms when warm and cold air masses collide and rotate. They're narrow, very fast, and short-lived — but the strongest can have wind speeds over 200 mph.

Question 35

WEATHER & BIOMES

Hurricanes form over:

A Cold ocean water

B Warm ocean water near the equator ✓ ANSWER

C Mountain ranges

D Polar ice sheets

SOLUTION

Hurricanes need warm ocean water (about 80 °F or warmer) for fuel. Warm water evaporates quickly, releasing heat and moisture that powers the storm. That's why hurricanes form near the equator and weaken over land or cold water.

Question 36

WEATHER & BIOMES

A long period of below-average rainfall is called a:

A Hurricane

B Drought ✓ ANSWER

C Flood

D Tornado

SOLUTION

A drought is a long stretch of unusually low precipitation. It can damage crops, dry up reservoirs, and worsen wildfires — serious problems in the Southwest.

Question 37

WEATHER & BIOMES

Which biome has the **highest biodiversity** (the most different species)?

A Tundra

B Desert

C Tropical rainforest ✓ ANSWER

D Taiga

SOLUTION

Tropical rainforests have warm temperatures year-round, abundant rain, and lots of sunlight — the perfect conditions for an enormous variety of plants and animals to thrive.

Question 38

WEATHER & BIOMES

The biome characterized by **permafrost** (permanently frozen soil) and short summers is the:

- A Tundra ✓ ANSWER
- B Tropical rainforest
- C Grassland
- D Desert

SOLUTION

The tundra is found near the Arctic Circle. Soil stays frozen year-round below the surface (permafrost), so trees can't take root. Plants are short and life is sparse.

Question 39

WEATHER & BIOMES

Cacti, kangaroo rats, and rattlesnakes are all adapted to which biome?

- A Tundra
- B Taiga
- C Desert ✓ ANSWER
- D Marine

SOLUTION

All three are classic desert species — very common right here around Vistancia! Cacti store water; kangaroo rats get water from their food; rattlesnakes hunt at night to avoid heat.

Question 40

WEATHER & BIOMES

The **vast majority** of Earth's water is found in the:

A Lakes and rivers

B Glaciers and ice caps

C Oceans ✓ ANSWER

D Atmosphere

SOLUTION

About 97% of Earth's water is salty ocean water. Glaciers/ice caps hold about 2%; only ~1% is the freshwater we can actually drink (rivers, lakes, groundwater).

Question 41

WEATHER & BIOMES

The amount of **solar energy** a region receives depends most on its:

A Time of day only

B Latitude (distance from the equator) ✓ ANSWER

C Wind speed

D Plant cover

SOLUTION

Near the equator, sunlight hits Earth almost straight on, concentrating energy. Near the poles, the same sunlight spreads over more area at a low angle, so it's much weaker. Latitude is the biggest driver of climate.

Question 42

WEATHER & BIOMES

In the **eye** of a hurricane, air pressure is:

A Higher than the surrounding air

B Equal to the surrounding air

C Lower than the surrounding air ✓ ANSWER

D Always exactly zero

SOLUTION

Hurricanes are intense low-pressure systems. The eye is the lowest-pressure part of the storm — that's why air rushes in toward it from all directions, fueling the spinning winds in the eyewall.

Question 43

WEATHER & BIOMES

Which biome typically has **4 distinct seasons** and trees that lose their leaves each fall (deciduous)?

A Tropical rainforest

B Temperate forest ✓ ANSWER

C Tundra

D Desert

SOLUTION

The temperate forest (much of the eastern U.S., Europe, parts of Asia) has cold winters, warm summers, and trees like oak and maple that drop their leaves to conserve water during the cold season.

Question 44

PLATE TECTONICS

The **thinnest, outermost** layer of Earth (where we live) is the:

A Mantle

B Outer core

C Inner core

D Crust ✓ ANSWER

SOLUTION

From the surface inward: crust → mantle → outer core → inner core. The crust is rocky and thin (5-70 km thick) compared to Earth's ~6,400 km radius.

Question 45

PLATE TECTONICS

Which layer of Earth is a **liquid** made mostly of melted iron and nickel?

A Inner core

B Outer core ✓ ANSWER

C Crust

D Lithosphere

SOLUTION

The outer core is liquid iron and nickel. The inner core is also iron/nickel but is solid because of extreme pressure. The motion of the liquid outer core generates Earth's magnetic field.

Question 46

PLATE TECTONICS

The scientist who first proposed that the continents had drifted apart over time was:

A Isaac Newton

B Alfred Wegener ✓ ANSWER

C Charles Darwin

D James Hutton

SOLUTION

Alfred Wegener (1912) noticed that continents like Africa and South America fit together like puzzle pieces and shared matching fossils and rocks. His idea was rejected at first because he couldn't explain HOW they moved — later confirmed by plate tectonics.

Question 47

PLATE TECTONICS

The supercontinent that existed about 250 million years ago, before breaking apart, was:

A Gondwana

B Laurasia

C Pangaea ✓ ANSWER

D Atlantis

SOLUTION

Pangaea (Greek for "all lands") broke apart into *Laurasia* (north) and *Gondwana* (south), which kept splitting until they formed today's continents. Atlantis is mythological.

Question 48

PLATE TECTONICS

Tectonic plates move because of:

A Earth's rotation alone

B The pull of the magnetic poles

C Convection currents in the mantle ✓ ANSWER

D Ocean tides on the surface

SOLUTION

Hot rock in the mantle rises, cools near the crust, sinks back down, and gets reheated — a cycle called a convection current. These slow currents drag the plates above them.

Question 49

PLATE TECTONICS

At a **divergent** plate boundary, plates are:

A Sliding past each other

B Moving apart from each other ✓ ANSWER

C Colliding head-on

D Completely still

SOLUTION

Memory tip: "di-VERGE" = "go in different directions" → moving apart. Magma rises into the gap, cools, and creates new crust.

Question 50

PLATE TECTONICS

A **mid-ocean ridge** forms at which type of plate boundary?

A Convergent (continent-continent)

B Divergent (oceanic) ✓ ANSWER

C Transform

D No boundary — it's a hot spot

SOLUTION

When two oceanic plates pull apart, magma rises and creates a long underwater mountain range — a mid-ocean ridge. The Mid-Atlantic Ridge is the most famous example. This is also where sea-floor spreading happens.

Question 51

PLATE TECTONICS

The **Himalayan mountains** formed when two continental plates collided. This is what type of boundary?

A Divergent

B Transform

C Convergent (continent-continent) ✓ ANSWER

D Hot spot

SOLUTION

When two continental plates collide, neither one can subduct (they're both light/thick). Instead the crust crumples upward into giant mountains. The Indian plate ramming into the Eurasian plate built the Himalayas, and they're still growing today.

Question 52

PLATE TECTONICS

The **San Andreas Fault** in California is a famous example of what kind of plate boundary?

A Convergent boundary

B Divergent boundary

C Transform boundary ✓ ANSWER

D Subduction zone

SOLUTION

At a transform boundary, plates slide horizontally past each other. They get stuck, build pressure, and snap — producing earthquakes. The Pacific Plate slides past the North American Plate along the San Andreas Fault.

Question 53

PLATE TECTONICS

When a denser **oceanic plate** meets a less dense **continental plate**, the oceanic plate:

A Slides on top of the continent

B Subducts (sinks below) the continental plate ✓ ANSWER

C Pushes the continent straight up

D Stops moving completely

SOLUTION

Oceanic crust is made mostly of basalt and is **denser** than the granite-rich continental crust. So the oceanic plate **subducts** down into the mantle, often forming a deep ocean trench and a chain of volcanoes inland.

Question 54

PLATE TECTONICS

The rigid outer shell of Earth (the crust + upper mantle), broken into tectonic plates, is called the:

A Asthenosphere

B Lithosphere ✓ ANSWER

C Hydrosphere

D Mesosphere

SOLUTION

"Litho-" means rock. The **lithosphere** is the solid rocky outer shell that's broken into the plates. Beneath it is the asthenosphere — hotter, partly soft mantle that the plates slide on.

Question 55

PERIODIC TABLE

The **atomic number** of an element equals the number of:

A Neutrons

B Electrons in the outermost shell

C Protons ✓ ANSWER

D Atomic mass in grams

SOLUTION

Atomic number = number of **protons**. The number of protons defines the element. (Hydrogen = 1, helium = 2, lithium = 3, and so on.) In a neutral atom, the number of electrons matches.

Question 56

PERIODIC TABLE

The **mass number** of an atom equals:

A Protons + electrons

B Protons + neutrons ✓ ANSWER

C Just the neutrons

D Protons × electrons

SOLUTION

Almost all of an atom's mass is in the nucleus, which holds **protons + neutrons**. Electrons are about 2,000 times lighter, so they don't really count toward mass. Subtract atomic # from mass # to get neutrons.

Question 57

PERIODIC TABLE

The horizontal **rows** on the periodic table are called:

A Groups

B Families

C Periods ✓ ANSWER

D Series

SOLUTION

Memory tip: "PERIOD = ROW" (think of a sentence ending — the period is at the end of a horizontal line). Periods tell you how many electron shells an element has.

Question 58

PERIODIC TABLE

The vertical **columns** on the periodic table are called:

A Periods

B Groups (or families) ✓ ANSWER

C Rows

D Levels

SOLUTION

Groups (families) are the columns. Elements in the same group share similar chemical properties because they have the same number of valence (outer) electrons.

Question 59

PERIODIC TABLE

The element with the symbol **Na** is:

A Nitrogen

B Sodium ✓ ANSWER

C Neon

D Nickel

SOLUTION

Na = sodium (from Latin *natrium*). Other Latin-based symbols to know: K = potassium (kalium), Fe = iron (ferrum), Au = gold (aurum), Ag = silver (argentum).

Question 60

PERIODIC TABLE

The **noble gases** (Group 18) are best described as:

A Highly reactive

B Inert (very unreactive) ✓ ANSWER

C Magnetic metals

D Liquid at room temperature

SOLUTION

Noble gases (helium, neon, argon, krypton, xenon, radon) have a "full" outer electron shell, so they don't need to react with other elements. Inert = doesn't easily react.

Question 61

PERIODIC TABLE

Which group contains the most **reactive metals**?

A Group 1 (alkali metals) ✓ ANSWER

B Group 18 (noble gases)

C Group 14

D Group 17 (halogens)

SOLUTION

Group 1 alkali metals (lithium, sodium, potassium…) have just one outer electron, which they "want" to lose. Sodium reacts violently with water; potassium is even worse. (Hydrogen sits in Group 1 but is not a metal.)

Question 62

PERIODIC TABLE

Which subatomic particle has **no electric charge**?

A Proton

B Electron

C Neutron ✓ ANSWER

D Photon

SOLUTION

Memory tip: NEUtron = NEUTRAL. Protons are +, electrons are −, neutrons have no charge (they balance the nucleus's mass without affecting charge).

Question 63

PERIODIC TABLE

Which element is a **metalloid** (has properties of both metals and nonmetals)?

A Iron (Fe)

B Silicon (Si) ✓ ANSWER

C Sodium (Na)

D Oxygen (O)

SOLUTION

Metalloids sit on the staircase line dividing metals from nonmetals (B, Si, Ge, As, Sb, Te). Silicon is the most famous — it's a semiconductor used in computer chips. Iron and sodium are metals; oxygen is a nonmetal.

Question 64

PERIODIC TABLE

An atom of **carbon (C)** has atomic number 6. How many **protons** does it have?

A 0

B 6 ✓ ANSWER

C 12

D 14

SOLUTION

Atomic number = number of protons. Carbon's atomic number is 6, so it has 6 protons. (Most carbon also has 6 neutrons, giving it a mass number of 12.)

Question 65

PERIODIC TABLE

Most of the elements on the periodic table are classified as:

A Nonmetals

B Noble gases

C Metals ✓ ANSWER

D Metalloids

SOLUTION

Roughly 75% of the periodic table is metals — everything to the left of the staircase line. Metals are typically shiny, conductive, malleable (bendy), and ductile (can be stretched into wire).

SECTION B · FILL-IN-THE-BLANK (66-75)

Question 66

FILL-IN · CELLS

The organelle that acts as the **"control center"** of a eukaryotic cell and contains the cell's DNA is the _____.

ACCEPTED ANSWER: **nucleus**

NOTE

Accept "nucleus" or "the nucleus." Do NOT accept "nucleolus" (that's a structure inside the nucleus that makes ribosomes).

Question 67

FILL-IN · PERIODIC TABLE

The chemical **symbol** for the element sodium is _____.

ACCEPTED ANSWER: **Na**

NOTE

Must be written with capital N and lowercase a (Na). From the Latin *natrium*. Do NOT accept "S," "So," or "Sd."

Question 68

FILL-IN · PLATE TECTONICS

The supercontinent that existed about 250 million years ago, before the modern continents drifted to where they are today, was called _____.

ACCEPTED ANSWER: **Pangaea**

NOTE

Accept "Pangaea," "Pangea," or any reasonable phonetic spelling. Greek for "all lands."

Question 69

FILL-IN · PERIODIC TABLE

An atom of oxygen has an atomic number of 8. The number of **protons** in the nucleus of this atom is _____.

ACCEPTED ANSWER: **8**

NOTE

Atomic number = number of protons, always. So oxygen with atomic number 8 has 8 protons.

Question 70

FILL-IN · BIOMES

A cold biome with permafrost (permanently frozen ground), low precipitation, and short summers, found near the Arctic Circle, is called the _____.

ACCEPTED ANSWER: **tundra**

NOTE

Accept "tundra" or "Arctic tundra." Do NOT accept "taiga" (that's the boreal forest just south of the tundra) or "tropical" anything.

Question 71

FILL-IN · PLATE TECTONICS

The type of plate boundary at which two tectonic plates pull **away** from each other is called a _____ boundary.

ACCEPTED ANSWER: **divergent**

NOTE

Accept "divergent" only. Do NOT accept "convergent" (which is plates colliding) or "transform" (sliding past).

Question 72

FILL-IN · BODY SYSTEMS

The body system that pumps blood through arteries, veins, and capillaries to carry oxygen, nutrients, and wastes is called the _____ system.

ACCEPTED ANSWER: **circulatory**

NOTE

Accept "circulatory" or "cardiovascular." Do NOT accept "respiratory" (that's lungs) or "vascular" by itself.

Question 73

FILL-IN · NON-CONTACT FORCES

The non-contact force that pulls every object with mass toward every other object with mass — including what holds the Moon in orbit around Earth — is _____.

ACCEPTED ANSWER: **gravity**

NOTE

Accept "gravity" or "gravitational force." Do NOT accept "magnetism" or "electric force."

Question 74

FILL-IN · CELLS

The process by which plants use sunlight, water, and carbon dioxide to make their own food (glucose) is called _____.

ACCEPTED ANSWER: **photosynthesis**

NOTE

Accept "photosynthesis." Do NOT accept "respiration" or "cellular respiration" (those break glucose down for energy — the opposite process).

Question 75

FILL-IN · PLATE TECTONICS

Wegener's idea that the continents have slowly moved over Earth's surface throughout history is called the theory of _____ _____ (two words).

ACCEPTED ANSWER: **continental drift**

NOTE

Accept "continental drift." Plate tectonics is the modern theory that *explains* continental drift, but Wegener's original term is "continental drift."

SECTION C · FREE RESPONSE (76-80)

Question 76

FRQ · CELLS

Compare and contrast a plant cell and an animal cell.

(a) List at least **three** organelles found in BOTH plant and animal cells, and briefly describe the job of each.

(b) List at least **two** structures found ONLY in plant cells and explain why a plant needs each one.

(c) Animal cells often look round or irregular while plant cells look more like rectangles or boxes. Explain *why* plant cells have this shape.

(A) ORGANELLES IN BOTH (ANY 3)

Nucleus — control center; stores DNA and directs the cell's activities.
Mitochondria — "powerhouse"; performs cellular respiration to release

energy from glucose.
Ribosomes — build proteins by reading instructions from the nucleus.
Cell membrane — controls what enters and leaves the cell.
Cytoplasm — jelly-like fluid that holds the organelles and is where many reactions happen.
Endoplasmic reticulum / Golgi apparatus — package and ship proteins.

(B) ONLY IN PLANT CELLS (ANY 2)

Cell wall — a rigid layer outside the cell membrane made of cellulose. It gives the plant structural support so the plant can stand up without bones.
Chloroplasts — contain chlorophyll and carry out photosynthesis. Plants need them because they make their own food using sunlight.
Large central vacuole — stores water, nutrients, and waste, and helps keep the plant cell firm by pushing outward against the cell wall.

(C) WHY PLANT CELLS ARE BOX-SHAPED

Plant cells have a **rigid cell wall** outside their cell membrane. The cell wall doesn't bend or shift much, so the cell holds a fixed, geometric box shape. Animal cells have only a flexible cell membrane (no wall), which is why they can be round, flat, or irregular and can change shape easily.

Question 77

FRQ · BODY SYSTEMS

Describe how the **respiratory system** and the **circulatory system** work TOGETHER to deliver oxygen from the air all the way to the cells in your body, and to remove carbon dioxide.

In your answer, you must use and correctly explain at least **five** of these terms: lungs, alveoli, diaphragm, red blood cells, hemoglobin, capillaries, heart, oxygen, carbon dioxide, mitochondria.

MODEL RESPONSE · PATH OF AN OXYGEN MOLECULE

Step 1. When the diaphragm contracts and flattens, the chest expands and air is pulled into the body. The air travels through the nose and trachea

into the **lungs**.

Step 2. Inside the lungs, the air ends up in tiny sacs called **alveoli**. Each alveolus is wrapped in tiny blood vessels called **capillaries**, and the walls of both are very thin. **Oxygen** diffuses out of the alveolus and into the blood.

Step 3. Oxygen attaches to **hemoglobin** inside **red blood cells**. The blood (now oxygen-rich) flows back to the **heart**, which pumps it out through arteries to the rest of the body.

Step 4. At each body cell, the blood travels through capillaries again. Oxygen diffuses out of the red blood cells, into the cells, and into the **mitochondria**. There, the oxygen is used in cellular respiration to release energy from glucose.

Step 5 (return trip). The waste gas **carbon dioxide** from cellular respiration enters the blood, travels back through the heart to the lungs, diffuses into the alveoli, and is exhaled when the diaphragm relaxes.

Big idea: The respiratory system handles *air ↔ blood* exchange. The circulatory system handles *blood ↔ cells* transport. Neither system can deliver oxygen to a cell on its own — they only work as a team.

Question 78

FRQ · PLATE TECTONICS

Compare the three main types of **plate boundaries**: convergent, divergent, and transform. For each, describe (a) what the plates are doing, (b) one geological feature or event that forms there, and (c) one real-world example.

CONVERGENT BOUNDARY

(a) Motion: Plates collide / push toward each other.
(b) Feature/event: Mountain ranges (continent-continent), volcanoes and ocean trenches (oceanic-continental subduction), island arcs (oceanic-

oceanic). Earthquakes are common.
(c) Example: Himalayas (India + Eurasia colliding); Andes Mountains and Cascade Range volcanoes (oceanic plates subducting under continental plates).

DIVERGENT BOUNDARY

(a) Motion: Plates move apart from each other.
(b) Feature/event: Mid-ocean ridges (under the sea) where new crust forms; rift valleys (on land); shallow earthquakes.
(c) Example: Mid-Atlantic Ridge (between the Americas and Europe/Africa); the East African Rift Valley.

TRANSFORM BOUNDARY

(a) Motion: Plates slide horizontally past each other in opposite directions.
(b) Feature/event: Faults and frequent earthquakes. (No volcanoes — crust is not being created or destroyed.)
(c) Example: San Andreas Fault in California, where the Pacific Plate slides past the North American Plate.

Question 79

FRQ · WEATHER & BIOMES

The **tropical rainforest** biome has very high biodiversity, while the **desert** biome has much lower biodiversity.

(a) Identify at least **three abiotic factors** that explain why the rainforest can support so many species.

(b) Identify at least **two abiotic factors** that make the desert harsh.

(c) Describe two **adaptations** — one rainforest, one desert — and explain how each helps the organism survive.

(A) WHY THE RAINFOREST HAS HIGH BIODIVERSITY (ANY 3)

Warm, stable temperatures year-round — no harsh winter to survive, so cold-blooded animals and tropical plants thrive all year.
Heavy, consistent rainfall (often 80+ inches a year) — abundant water for plants and the animals that feed on them.
Plenty of sunlight near the equator — powers photosynthesis and supports a thick, layered canopy.
Nutrient-rich, layered habitat (forest floor, understory, canopy, emergent layer) — many different niches for different species to live in.

(B) WHY THE DESERT IS HARSH (ANY 2)

Very low precipitation — less than 10 inches per year. Water is the limiting resource for life.
Extreme temperatures — very hot during the day, often very cold at night, which is hard on living things.
Intense sunlight and high evaporation — what little water exists evaporates quickly.
Poor, sandy soil — few nutrients for plants.

(C) TWO ADAPTATIONS

Rainforest example — sloths: Sloths move very slowly and have algae-tinted fur, which camouflages them from predators in the green canopy. Their slow metabolism also lets them survive on a low-energy diet of leaves.

Desert example — cacti: Cacti have thick, waxy stems that store water and a small surface area (with spines instead of leaves) that reduces water loss. Their shallow, wide roots quickly soak up rain when it does fall. These adaptations let cacti survive months without rain.

Question 80

FRQ · PERIODIC TABLE + FORCES

An atom of **magnesium (Mg)** has an atomic number of **12** and a mass number of **24**.

(a) How many **protons**, **neutrons**, and **electrons** does a ***neutral*** magnesium atom have?

(b) In which **group** and **period** of the periodic table is magnesium located?

(c) List **two** general physical properties of metals.

(d) Two magnesium ions (each **+2**) are placed near each other. What kind of **non-contact force** exists between them, and is it attractive or repulsive? Why?

(A) PROTONS, NEUTRONS, ELECTRONS

Protons = atomic number = 12.
Neutrons = mass number − atomic number = 24 − 12 = 12.
Electrons = 12 (a *neutral* atom has equal numbers of protons and electrons so the +12 charge from protons cancels with the −12 charge from electrons).

(B) LOCATION ON PERIODIC TABLE

Group 2 (alkaline earth metals) and Period 3. Mg sits right below beryllium (Be) and right above calcium (Ca).

(C) TWO PROPERTIES OF METALS (ANY 2)

Shiny / lustrous when polished.
Good conductors of heat and electricity.
Malleable (can be hammered into sheets).
Ductile (can be drawn into wires).
Solid at room temperature (except mercury).

(D) FORCE BETWEEN TWO MG^{2+} IONS

Both ions have a charge of +2. According to the rule "like charges repel," the non-contact force between them is the electric force (electrostatic force), and it is repulsive — the ions push away from each other.

Reasoning: Two positive charges have the same sign, so they repel. (If one were +2 and the other −2, the force would be attractive instead.)